CA2 Suggestions- MIM501- Financial Management

(Module-1,3,4,5)

CA2 Question Paper Pattern [25 Marks]:

Short Question (1 mark x 5 Ques.)

Long Question (5 mark x 4 Ques.)

NOTE: PLEASE BRING A CALCULATOR TO THE EXAM

Section A: (5x1)

Module 1 – Introduction to Financial Management

1. Wealth maximization considers the time value of money, while profit maximization does not. **(True/False)**

2. Which of the following is NOT one of the three basic decisions in Financial Management?
   a) Investment Decision b) Financing Decision c) Dividend Decision d) Taxation Decision

3. The minimum required rate of return a firm must earn to satisfy its investors/lenders is called _____.
   Answer: Cost of Capital

4. What does PVIF stand for in financial management?
   (A) Present Value Interest Factor (B) Present Value Investment Factor (C) Principal Value Interest Factor

5. Cost of debt cannot be negative in general- true or false

Module 3 & 5 – Capital Budgeting & Investment Decisions

6. State the decision rule for accepting a project under NPV.
   Answer: Accept the project if NPV > 0; reject if NPV < 0.

7. If PI (Profitability Index) of a project is 1.2, the project should be:
   a) Rejected b) Accepted c) Held for review d) Cannot be determined

8. Payback Period considers the time value of money. **(True/False)**

9. The discount rate at which NPV of a project becomes zero is called _____.
   Answer: IRR (Internal Rate of Return)

10. Write the formula for Payback Period (for even cash inflows).
    Answer: PBP = Initial Investment / Annual Cash Inflow

Module 4 – Leverage Analysis & Capital Structure Theories

11. A high Degree of Financial Leverage (DFL) indicates:
    a) High fixed operating costs b) More use of debt, riskier for shareholders if earnings fall c) Stable dividend policy d) Low business risk

12. According to the Net Operating Income (NOI) Approach, the overall cost of capital remains constant regardless of the debt-equity mix. **(True/False)**

13. The type of leverage that arises due to fixed financial costs like interest is called _____ leverage.
    Answer: Financial

14. If DOL = 2 and DFL = 1.5, what is the Degree of Combined Leverage (DCL), and what does it mean for a 1% change in sales?
Answer: DCL = DOL × DFL = 3; a 1% change in sales leads to a 3% change in EPS.

Section B: (4x5)

MODULE 1: INTRODUCTION TO FINANCIAL MANAGEMENT-

1. Discuss how the cost of capital enters into the process of evaluating capital budgeting proposals.

Ans: Capital budgeting involves long-term investment appraisal, and the **cost of capital** serves as the **cut-off rate** for project selection.
Role in Capital Budgeting:
• **Net Present Value (NPV):** Future cash inflows are discounted at the firm's cost of capital. A project is accepted if NPV > 0.
• **Internal Rate of Return (IRR):** A project is viable if IRR exceeds the cost of capital.
• **Profitability Index (PI):** Ratio of present value of inflows to outflows, using cost of capital as the discount rate.

Importance:
• Ensures projects earn more than the financing cost.
• Reflects time value of money and investor expectations.
• Aligns investment decisions with shareholder wealth maximization.

Example: If a company's Weighted Average Cost of Capital (WACC) is 12% and a project's IRR is 15%, the project should be accepted.
Conclusion: Cost of capital acts as the benchmark for discounted cash flow methods and ensures only value-adding projects are selected.

2. A company issued 10,000, 10% debentures of ₹ 100 each at a premium of 10% on 1.4.2023 to be matured on 1.4.2028. The debentures will be redeemed at par on maturity. COMPUTE the cost of debentures assuming 35% as tax rate.

Ans: The cost of debenture (K_d) will be calculated as below:

$$\text{Cost of debenture } (K_d) = \frac{I(1-t) + \frac{(RV-NP)}{n}}{\frac{(RV+NP)}{2}}$$

I= Interest on debenture= 10% of Rs.100 = ₹100
NP= Net Proceeds = 110% of Rs 100= ₹110
RV= Redemption value= ₹100
n = Period of debenture = 5yrs
t = Tax rate= 35% or 0.35

$$K_d = \frac{₹10(1-0.35) + \frac{(₹100-₹110)}{5 \text{ years}}}{\frac{(₹100+₹110)}{2}}$$

$$\text{Or, } K_d = \frac{(₹10 \times 0.65) - ₹2}{₹105} = \frac{₹4.5}{₹105} = 0.0428 \text{ or } 4.28\%$$

3. Asianol Ltd. has the following Capital Structure:

Source of Capital	₹ (in Lakhs)
Equity Share Capital (10 lakhs shares)	100
12% Preference Share Capital (10,000 shares)	10
Retained Earnings	120
14% Debentures (70,000 Debentures)	70
14% Term Loan	100
Total	400

The Market price per equity share is ₹25. The next expected dividend per share is ₹2 and is expected to grow at 8%. The preference shares are redeemable after 7 years at par and are currently quoted at ₹75 per share. Debentures are redeemable after 6 years at par and their current market quotation is ₹90 per debenture. The tax rate applicable to the firm is 50%. Find: i) K_e and K_r ii) K_p iii)K_d iv)K_{tl}

Ans: i) Cost of Equity- $K_e = \frac{D_1}{P_0} + g$

$D_1 = ₹2, P_0 = ₹25, g = 0.08$

$K_e = \frac{2}{25} + 0.08 = 0.08 + 0.08 = 0.16$ or **16%**

Since no flotation costs are given, Cost of Retained Earnings $K_r = K_e$= 16%

ii) Cost of Preference Share- $K_p = \frac{D_p + \frac{RV-NP}{n}}{\frac{RV+NP}{2}}$

$D_p = 12\%$ of ₹100 = ₹12, $RV = ₹100, NP = ₹75, n = 7$ years

$K_p = \frac{12 + \frac{100-75}{7}}{\frac{100+75}{2}} = \frac{12 + 3.57}{87.5} = \frac{15.57}{87.5} = 0.1779$ or **17.79%**

iii) Cost of Debt(debentures)- $K_d = \frac{I(1-t) + \frac{RV-NP}{n}}{\frac{RV+NP}{2}}$

$I = 14\%$ of ₹100 = ₹14, $t = 0.50, RV = ₹100, NP = ₹90, n = 6$ years

$K_d = \frac{14(1-0.50) + \frac{100-90}{6}}{\frac{100+90}{2}} = \frac{7 + 1.67}{95} = \frac{8.67}{95} = 0.0913$ or **9.13%**

iv) Cost of Term Loan-

$K_{tl} = I(1-t)$

$I = 14\%, t = 0.50$

$K_{tl} = 14\% \times (1 - 0.50) = 7\%$

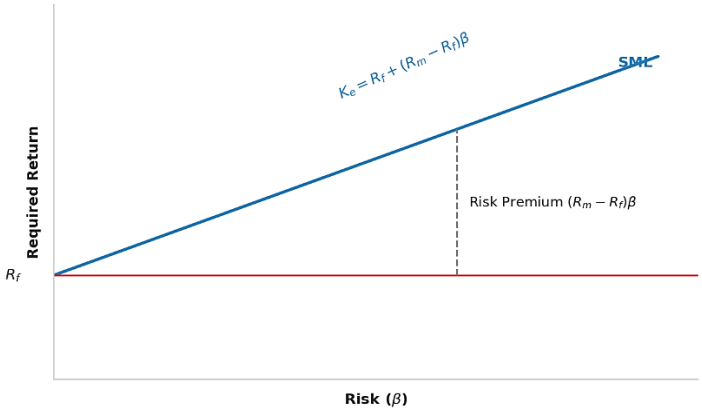
4. Write a short note on CAPM(Capital Asset Pricing Model).

Ans: Capital Asset Pricing Model (CAPM) is a financial model developed by William F. Sharpe, John Lintner, and Jan Mossin in the 1960s. It explains the relationship between the expected return and systematic risk of an investment and is widely used to estimate a firm's cost of equity.

The CAPM formula is:

$$K_e = R_f + \beta(R_m - R_f)$$

Where:
K_e = Expected return or cost of equity
R_f= Risk-free rate of return
β = Beta, measuring systematic risk
R_m= Expected market return



Types of Risk under CAPM

Investment risk can broadly be classified into two categories:

1. Systematic Risk
This is the risk arising from factors that affect the overall market or economy, such as inflation, changes in interest rates, recession, or economic policies. It cannot be eliminated through diversification and is measured by Beta (β).

2. Unsystematic Risk
This risk arises from factors specific to a particular company or industry, such as poor management, strikes, product failure, or legal problems. It can be reduced or eliminated through diversification and therefore is not separately rewarded under CAPM.

Module 3 & 5 – Capital Budgeting & Investment Decisions

1. Internal Rate of Return (IRR) of a project is that rate where Net Present Value (NPV) is zero. Comment.

Ans: The Internal Rate of Return (IRR) of a capital budgeting project is the discount rate at which the Net Present Value (NPV) of a project **equals zero**. The IRR decision rule specifies that all independent projects with an IRR greater than the cost of capital should be accepted. When choosing among mutually exclusive projects, the project with the highest IRR should be selected (as long as the IRR is greater than the cost of capital).

Acceptance rule: The use of IRR, as a criterion to accept capital investment decisions, involves a comparison of IRR with the required rate of return known as cut-off-rate. If IRR is greater than cut-off-rate, the project should be accepted. If IRR is less than the cut-off rate, the project is rejected. In case, the IRR is equal to the cut-off-rate, the firm is indifferent.

Merits:
(1) It considers the time value of money.
(2) It takes into account the total cash inflows and cash outflows.
(3) It is easier to understand

Demerits:
(1) It does not use the concept of desired rate of return, whereas it provides the rate of return, which is indicative of the profitability of investment proposal.
(2) It involves tedious calculations, based on trial and error method.
(3) It produces multiple rates, which can be confusing.
(4) Projects selected based on higher IRR may not be profitable.
(5) Unless the life of the project can be accurately estimated, assessment of cash flows cannot be correctly made.
(6) Single discount rate ignores the varying future interest rates.

2. Payback period is suitable to measure liquidity of a project rather than its profitability. Comment.

Ans: The payback period represents the amount of time that it takes for a capital budgeting project to recover its initial cost. The use of the payback period as a capital budgeting decision rule specifies that all *independent* projects with a payback period less than a specified number of years should be accepted. When choosing among *mutually exclusive* projects, the project with the quickest payback is preferred.
Merits:
(1) It is simple to apply, easy to understand.
(2) In case of capital rationing, a company is compelled to invest in projects having shortest payback period.
(3) This method is most suitable when the future is very uncertain. The shorter the payback period, the less risky is the project. Therefore, it can be considered as an indicator of risk.
(4) This method gives an indication to the prospective investors specifying when their funds are likely to be repaid.
(5) It does not involve assumptions about future interest rates.
(6) Ranking projects according to their ability to repay quickly may be useful to firms when experiencing liquidity constraints. They will need to exercise careful control over cash requirements.
Demerits:
(1) It does not indicate whether an investment should be accepted or rejected, unless the Payback period is compared with an arbitrary managerial target.
(2) The method ignores cash generation beyond the payback period and this can be seen more a measure of liquidity than of profitability.
(3) It fails to take into account the timing of returns and the cost of capital.
(4) The traditional payback approach does not consider the salvage value of an investment. It fails to determine the payback period required in order, to recover the initial outlay if things go wrong.
(5) This method makes no attempt to measure a percentage return on the capital invested and is often used in conjunction with other methods.

3. **ABC Ltd** is evaluating the purchase of a new machinery with a depreciable base of **₹ 1,00,000**; expected economic life of **4 years** and change in earnings before taxes and depreciation of **₹ 45,000 in year 1, ₹ 30,000 in year 2, ₹ 25,000 in year 3 and ₹ 35,000 in year 4**. Assume **straight-line depreciation** and a **20% tax rate**. You are required to **COMPUTE relevant cash flows**.

Ans: Depreciation = ₹ 1,00,000 ÷ 4 = ₹ 25,000

Amount in (₹)

Particulars	Years			
	1	**2**	**3**	**4**
Earnings before tax and depreciation	45,000	30,000	25,000	35,000
Less: Depreciation	(25,000)	(25,000)	(25,000)	(25,000)
Earnings before tax	20,000	5,000	0	10,000
Less: Tax @20%	(4,000)	(1,000)	0	(2,000)
Earnings after tax	16,000	4,000	0	8,000
Add: Depreciation	25,000	25,000	25,000	25,000
Net Cash flow	**41,000**	**29,000**	**25,000**	**33,000**

4. A project requiring an investment of ₹ 10,00,000 and it yields profit after tax and depreciation which is as follows:

Years	Profit after tax and depreciation (₹)
1	50,000
2	75,000
3	1,25,000
4	1,30,000
5	80,000
Total	**4,60,000**

Suppose further that at the end of the 5th year, the plant and machinery of the project can be sold for **₹ 80,000**. **DETERMINE Average Rate of Return**.

Ans: (a) If Initial Investment is considered then,

$$\frac{\text{Total Profit} \div \text{No. of years}}{\text{Average investment/Initial Investment}} \times 100$$

$$= \frac{₹\,4{,}60{,}000 \div 5 \text{ years}}{₹\,10{,}00{,}000} \times 100 = \frac{₹\,92{,}000}{₹\,10{,}00{,}000} \times 100 = \mathbf{9.2\%}$$

(b) If Average investment is considered, then,

$$= \frac{₹\,92{,}000}{\text{Average Investment}} \times 100 = \frac{₹\,92{,}000}{₹\,5{,}40{,}000} \times 100 = \mathbf{17.04\%}$$

Where,

$$\text{Average Investment} = \frac{1}{2}(\text{Initial investment} - \text{Salvage value}) + \text{Salvage value}$$
$$= \frac{1}{2}(₹\,10{,}00{,}000 - ₹\,80{,}000) + ₹\,80{,}000$$
$$= ₹\,4{,}60{,}000 + ₹\,80{,}000 = ₹\,5{,}40{,}000$$

[**Note:** ***Initial Investment assumes your capital is fully tied up for the entire 5 years. It is conservative and tends to understate returns.***

Average Investment accounts for the fact that asset value depreciates over time. Since the book value of the machinery decreases every year down to its salvage value, your "average" locked capital is much lower, which naturally pushes the calculated rate of return upward.]

MODULE 4: LEVERAGES

1. A firm's sales, variable costs and fixed cost amount to ₹ 75 lakh, ₹ 42 lakh and ₹ 6 lakh respectively. It has borrowed ₹ 45 lakh at 9% and its equity capital totals ₹ 55 lakh.
(a) What is the firm's ROI?
(b) Does it have favorable financial leverage?
(c) If the firm belongs to an industry whose asset turnover is 3, does it have high or low asset leverage?
(d) What are the operating, financial and combined leverage of the firm?

Ans:

(a) ROI = EBIT/Investment

EBIT = Sales – VC – FC

= ₹ 75 lakh – ₹ 42 lakh – ₹ 6 lakh

= ₹ 27 lakh

ROI = ₹ 27 lakh/₹ 100 lakh

= 27 %

(b) Yes, the firm has favourable financial leverage as its ROI is higher than the interest on debt.

(c) Asset turnover = Sales/Total Assets or Total Investments = ₹ 75 lakh/₹ 100 lakh = 0.75. It is lower than the industry average.

(d) Operating Leverage $= \frac{\text{Sales - Variable costs}}{\text{EBIT}} = \frac{₹\ 75\ \text{lakh} - ₹\ 42\ \text{lakh}}{₹\ 27\ \text{lakh}} = 1.22$

Financial Leverage $= \frac{\text{EBIT}}{\text{EBIT - Interest}} = \frac{₹\ 27\ \text{lakh}}{₹\ 27\ \text{lakh} - ₹\ 4.05\ \text{lakh}} = 1.18$

Combined Leverage $= \frac{\text{Sales - VC}}{\text{EBIT - Interest}} = \frac{₹\ 33\ \text{lakh}}{₹\ 22,95,000} = 1.44$

Alternatively, $= \text{OL} \times \text{FL} = 1.22 \times 1.18 = 1.44$

2. Calculate the operating leverage for each of the four firms, A, B, C and D from the following price and cost data. What conclusions can you draw with respect to levels of fixed cost and the degree of operating leverage DOL result? Explain. Assume number of units sold is 5,000.

Particulars	Firm A	Firm B	Firm C	Firm D
Sale Price per Unit (₹)	20	32	50	70
Variable Cost per Unit (₹)	6	16	20	50
Fixed Operating Cost (₹)	80,000	40,000	2,00,000	Nil

Ans:

Particulars	Firm A (₹)	Firm B (₹)	Firm C (₹)	Firm D (₹)
Sales (Units)	5,000	5,000	5,000	5,000
Sales revenue (Units × cross ×Price)	1,00,000	1,60,000	2,50,000	3,50,000
Less: Variable cost (Units × cross × VC per unit)	30,000	80,000	1,00,000	2,50,000
Contribution	70,000	80,000	100,000	100,000
Less: Fixed Operating Costs	80,000	40,000	2,00,000	Nil
EBIT	**(10,000)**	**40,000**	**(50,000)**	**1,00,000**

$$DOL = \frac{\text{Current Sales (S) - Variable Costs (VC)}}{\text{Current EBIT}}$$

$$DOL_{(A)} = \frac{₹\,1,00,000 - ₹\,30,000}{₹\,10,000}$$

$$= 7$$

$$DOL_{(B)} = \frac{₹\,1,60,000 - ₹\,80,000}{₹\,40,000}$$

$$= 2$$

$$DOL_{(C)} = \frac{₹\,2,50,000 - ₹\,1,00,000}{₹\,50,000}$$

$$= 3$$

$$DOL_{(D)} = \frac{₹\,3,50,000 - ₹\,2,50,000}{₹\,1,00,000}$$

$$= 1$$

Conclusion

- Firms with higher fixed operating costs (Firm C: ₹2,00,000; Firm A: ₹80,000) show negative EBIT at this sales level, resulting in negative DOL — indicating they are operating below their break-even point and carry the highest business risk.
- Firm D, with no fixed cost, shows DOL = 1.00 — EBIT moves exactly in proportion to sales, representing the lowest business risk.
- Firm B, with moderate fixed cost (₹40,000), shows a stable DOL of 2.00 with positive EBIT.

Overall: The higher the fixed operating cost, the higher the sensitivity of EBIT to sales (higher DOL) — and hence, higher business risk. Firms with low/no fixed costs remain more stable and profitable even at lower sales volumes, while firms with high fixed costs need a much larger sales volume to break even and avoid losses.

3. Elucidate: Financial Leverage is also called Trading on Equity.

Ans: Financial Leverage refers to the use of fixed-cost funds (debentures, long-term loans, preference capital) along with equity capital in a firm's capital structure. It measures the effect of fixed financial charges (interest) on the Earnings Per Share (EPS) of equity shareholders.

Trading on Equity means using debt capital, with the equity base as support, to increase the return available to equity shareholders. Here, "equity" refers to shareholders' funds, and "trading" means "making use of."

Reasons why Financial Leverage is called Trading on Equity:

i. Debt carries a fixed rate of interest, payable irrespective of the firm's profit level.

ii. When the firm's rate of return (ROI) is higher than the interest rate paid on debt, the surplus return does not go to lenders — it belongs entirely to equity shareholders, increasing EPS.
iii. The firm uses its equity base as security to raise debt, and channels the benefit of that debt back to equity shareholders.
iv. Financial Leverage (DFL) quantifies this exact effect — the magnified change in EPS due to a change in EBIT.

Formula: DFL = EBIT / EBT

Example

EBIT = ₹1,20,000; Interest = ₹20,000
EBT = 1,20,000 − 20,000 = ₹1,00,000
DFL = 1,20,000 / 1,00,000 = 1.2

(A 1% change in EBIT causes a 1.2% change in EPS.)

Financial Leverage uses fixed-cost debt to increase equity shareholders' earnings — this is the same financial phenomenon described by Trading on Equity.

4. What factors from the Income Statement affect or influence Business Risk and Financial Risk?

Ans: A. Factors affecting Business Risk (Sales → Contribution → EBIT)

Business risk arises from variability in EBIT due to fixed **operating** costs, measured by Operating Leverage (DOL).

1. **Sales Volume Fluctuation** – Instability in sales quantity affects Contribution and EBIT.
2. **Selling Price Variation** – Changes in price affect Contribution directly.
3. **Fixed Operating Costs** – Rent, depreciation, fixed salaries — higher fixed costs mean higher DOL and higher business risk.
4. **Variable Cost Ratio** – Higher variable costs relative to sales reduce the Contribution margin, affecting EBIT stability.
5. **Nature of Industry/Product** – Demand elasticity and cost structure differ by industry, affecting sensitivity of EBIT to sales changes.

(DOL = Contribution / EBIT)

B. Factors affecting Financial Risk (EBIT → EBT → EAT)

Financial risk arises from variability in EPS due to fixed **financial** costs, measured by Financial Leverage (DFL).

1. **Interest on Debt** – Fixed interest obligation on debentures/loans, payable regardless of profit level.
2. **Preference Dividend** – A fixed commitment, relevant when assessing overall fixed financial burden.
3. **Level of EBIT** – A low or unstable EBIT relative to fixed interest increases financial risk.
4. **Proportion of Debt in Capital Structure** – Higher debt means higher fixed interest, higher DFL, and higher financial risk.
5. **Tax Rate** – Affects the after-tax cost of interest (interest is tax-deductible, reducing effective burden).

(Formula link: DFL = EBIT / EBT)